

The AI Diffusion Pathway Framework

This framework serves two purposes: **(1)** generating a structured pathway document from raw source material, and **(2)** guiding an adopter conversationally toward relevant know-how. Both uses draw on the same underlying structure - dimensions, sub-categories, stages, unit types, and insight forms - just applied in different directions: generation *extracts* units from raw text into this structure; adopter guidance *retrieves* units from this structure to answer a live question.

1. The core structure

The four dimensions

Dimension	Central question	Sub-categories
Persona	Are we solving the right problem for the right person?	A. Problem and Persona · B. Current Journey and Friction · C. Outcome and Success · D. Scope, Inclusion, and Trust
Solution	Are we building the right system to solve it?	A. AI Fit and Comparative Advantage · B. UX, Channel, and Integration · C. Model, Architecture, and Infrastructure · D. Data and Knowledge Readiness · E. Performance, Reliability, and Scale
Institution	Can the institution own, absorb, govern, and sustain it?	A. Mandate, Ownership, and Decision Rights · B. Workforce and Change · C. Governance, Safety, and Redress · D. Accountability, Liability, and Compliance · E. Data Stewardship · F. Operating Model and Sustainability · G. Institutionalisation and Continuous Improvement
Ecosystem	Can the required network of actors execute and support it?	A. Partner Architecture and Roles · B. External Data and Infrastructure Dependencies · C. Delivery, Distribution, and Trust · D. Coordination, Procurement, and Incentives · E. Resilience, Portability, and Contingencies · F. Ecosystem Learning and Diffusion

30/70 thesis: Persona + Solution = defining and building the right thing. Institution + Ecosystem = the larger work of enabling adoption, accountability, and sustainability. Not four equal shares of effort.

Cross-cutting concerns

Data:

- **Persona** - evidence for problem/population/outcome
- **Solution** - is data usable/reliable/current
- **Institution** - who owns/authorises/corrects it), **Ecosystem** (external sources, agreements, SLAs

Trust:

- **Persona** - does the excluded user trust this channel enough to use it
- **Solution** - is trust conveyed through channel/UX choices - tone, disclosure, voice vs. text
- **Institution** - accountability. who the user blames when it's wrong, who stands behind the answer
- **Ecosystem** - trust in delivery partners and distribution channels, independent of the institution itself

The four adopter stages

Stage	Central question	Done when...
Explore	Is AI appropriate, and what would it take?	Precise excluded-user definition. Honest comparison with alternatives. Order-of-magnitude cost sense.
Define	What must be true before building?	Named data owners. Named mandate holder. Architecture posture chosen. Safety boundaries designed.
Pilot	What breaks with real users and real institutional conditions?	Failure taxonomy. Named institutional response to first public failure. Real cost-per-interaction data.
Scale	Can the institution own, sustain, and continuously improve it?	Budget line. Named operational owner. Monitoring mechanism. Operating model written down.

The five unit (micro-innovation) types

Type	Definition	What makes it reusable
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Strategic Decision	A framing, governance, or design decision that shaped what got built. Usually invisible in the final product.	The condition tag - when does this apply, when doesn't it?
Tactical Decision	A stack, sequence, cost, or implementation decision specific enough to reuse.	A before→after: what changed because of this decision.
Failure and Fix	Something that broke, the fix, and what the fix revealed about the system.	The fix reveals the structural insight - not the failure alone.
Playbook	A genuine multi-step, gated sequence for a recurring situation.	Actionable: "if X, do Y before Z."
Toolkit Asset	A reusable technical component, template, or governance artefact.	Another adopter can lift and adapt it without rebuilding.

Tag every unit: Dimension + Sub-category + Stage + Type + Condition tag.

2. The question bank

EXPLORE - Is AI the right answer, and what would it take?

Dimension	Core question	Listening for	Insight form	Corpus example
Persona	Who specifically is excluded from this service today - and what do they do instead?	The workaround is the data. It reveals the real replacement baseline.	Excluded user + named barrier + current workaround	MahaVISTAAR: women farmers receiving contradictory advice from fertiliser sellers. No trusted official source in their language.
Solution	What channel or system are you replacing, what does it fail at, and is AI actually the	Two human callers, paper forms, nothing - plus honest reasoning on	Current channel + failure mode + AI-fit justification	Lend A Hand: two human callers couldn't collect fortnightly feedback from thousands of dispersed interns.

	right tool for that failure?	AI-fit vs simpler fixes.		
Institution	Who inside the institution has to personally want this to work - and do they know yet?	Not procurement sign-off. The person whose professional stake is tied to success.	Named champion + their specific stake	MahaVISTAAR: Commissioner-level ownership required before any department would authorise data connections.
Ecosystem	Who else has tried to solve this for this population, and what happened?	Precedent deployments, failed attempts, adjacent tools. Who do they trust as a reference peer?	Named precedent + what transferred + what didn't	Ethiopia ATI drew on MahaVISTAAR: 9 months → 3 months. Architecture transferred. Farmer-trust mechanics needed local adaptation.

DEFINE - What are the irreversible decisions?

Dimension	Core question	Listening for	Insight form	Corpus example
Persona	What is the one question a user will ask that this system must answer, or the pilot fails?	Forces scope to its minimum. Shapes data model, prompt design, safety boundaries.	Single critical use case + binary success definition	MahaVISTAAR: "What should I spray this week in my village?" Required live mandi data, weather API, pest calendar, conversational Marathi.

Solution	Which architecture choices, if wrong, would take six months to undo - and is every data source named, with an accountable owner for each?	Residency, vendor lock-in, sovereignty, orchestration ownership. Not "government data" - ICAR, IMD, APMC, named humans.	Irreversible decisions list + data source registry (source × owner × cadence × accountability)	MahaVISTAAR: four named data owners before launch, each connected via API - AI layer consumes but doesn't own the data.
Institution	Who approves what the system says, have they agreed to own that, and what testing/timeline has the institution committed to before real users?	Content authority, not technical sign-off. Staged testing with named gate criteria.	Content authority + approval process + testing progression	Voice AI: the bot speaks in the department's name. The department must own every answer - including wrong ones.
Ecosystem	Which parts can you not build yourselves - and do you have a named partner for each?	Data owners, language models, telephony partners, integrators, field networks. Unnamed dependencies are unmanaged risk.	Dependency map: component × build-or-source × named partner	Bhili: eight distinct ecosystem roles, all named before model work. Project stalled when the linguist role was unnamed for three months.

PILOT - Live with real users. What's breaking?

Dimension	Core question	Listening for	Insight form	Corpus example
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Persona	Which user interactions are failing - is that a scope problem or a quality problem?	Scope = outside mandate. Quality = mandate questions answered badly. Different fixes.	Failure taxonomy: scope vs quality vs experience	Voice AI: out-of-mandate answers = institutional boundary crossing. Required explicit refusal design, not better prompting.
Solution	Which component is causing the most pain, is it replaceable, and which data source is going stale or wrong?	Bundled-vs-unbundled plays out here. Data quality failures at Pilot look like AI failures to users.	Component failure + replaceability + data quality issue + owner response time	MahaVISTAAR: mandi price 48hr lag caused wrong harvest-timing advice. Fix: governance call with APMC to 6-hr cadence - not a technical fix.
Institution	What has the institution seen fail publicly - and did they own it or disown it?	How the institution responds to the first failure is the strongest signal about whether this scales.	First public failure + institutional response (own vs disown)	Voice AI: bot answered outside mandate. Department tightened refusal boundaries - didn't shut down. Ownership held.
Ecosystem	Which partner is underperforming - and do you have an alternative?	Mid-pilot switching is painful but possible. Post-scale switching	Partner performance log + contingency plan	Voice AI: parallel vendor testing recommended precisely because mid-pilot switching is possible.

		requires a rebuild.		
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SCALE - Expanding to population. Can the institution own this without the founding team?

Dimension	Core question	Listening for	Insight form	Corpus example
Persona	Are new user segments arriving that the pilot wasn't designed for?	Scale reveals "the user" was multiple users - feature phone vs smartphone, dialect A vs B.	User segment expansion map + design change required per segment	Voice AI: multilingual demand discovered post-launch. Retrofit significantly harder than designing for it. Preventable.
Solution	Which components are you now unbundling, what triggered it - and which data sources are breaking under scale, do formal SLAs exist?	Cost/control arguments become concrete numbers at scale.	Unbundling decision (component × trigger × gain) + data SLA map	Bharat-VISTAAR: required formal data SLAs with 12 state agriculture departments. Each became an accountable node.
Institution	Has the institution absorbed this - budget line, named owner, review cadence - and does the system leave people more capable or more dependent?	Absorption vs "still the project team's problem." Workforce agency outcome.	Absorption indicators (budget + owner + review cadence) + agency outcome (capability vs dependency)	Voice AI: shift from "project" to "service" is the signal. Departmental adoption requires owning voice as a service.

Ecosystem	What from your deployment could the next adopter reuse - with what conditions?	Not "we did X." "X works when Y is true, fails when Z is true."	Transferable unit + condition tag (applies when / fails when)	MahaVISTAAR → Ethiopia ATI: architecture and trust-framing transferred. Specific data partnerships didn't - condition: government credibility as trust mechanism.
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Persona

Sub-category	Explore	Define	Pilot	Scale
A. Problem and Persona	Primary	Secondary	Dormant	Secondary
B. Current Journey and Friction	Primary	Secondary	Dormant	Dormant
C. Outcome and Success	Secondary	Primary	Primary	Primary
D. Scope, Inclusion, and Trust	Secondary	Primary	Primary	Primary

Solution

Sub-category	Explore	Define	Pilot	Scale
A. AI Fit and Comparative Advantage	Primary	Secondary	Dormant	Dormant

B. UX, Channel, and Integration	Secondary	Primary	Primary	Secondary
C. Model, Architecture, and Infrastructure	Secondary	Primary	Primary	Primary
D. Data and Knowledge Readiness	Secondary	Primary	Primary	Primary
E. Performance, Reliability, and Scale	Dormant	Secondary	Primary	Primary

Institution

Sub-category	Explore	Define	Pilot	Scale
A. Mandate, Ownership, and Decision Rights	Primary	Primary	Secondary	Secondary
B. Workforce and Change	Secondary	Primary	Primary	Primary
C. Governance, Safety, and Redress	Dormant	Primary	Primary	Secondary
D. Accountability, Liability, and Compliance	Dormant	Secondary	Secondary	Primary
E. Data Stewardship	Dormant	Primary	Secondary	Primary

F. Operating Model and Sustainability	Secondary	Primary	Secondary	Primary
G. Institutionalisation and Continuous Improvement	Dormant	Dormant	Secondary	Primary

Ecosystem

Sub-category	Explore	Define	Pilot	Scale
A. Partner Architecture and Roles	Secondary	Primary	Secondary	Secondary
B. External Data and Infrastructure Dependencies	Secondary	Primary	Secondary	Primary
C. Delivery, Distribution, and Trust	Dormant	Secondary	Secondary	Secondary
D. Coordination, Procurement, and Incentives	Dormant	Secondary	Secondary	Secondary
E. Resilience, Portability, and Contingencies	Dormant	Dormant	Primary	Primary
F. Ecosystem Learning and Diffusion	Dormant	Dormant	Secondary	Primary

Using it for generation: check raw content against the relevant cell's insight form before tagging a unit - not by how the content *sounds*, but by whether it actually contains the insight form's

components. Cross-check the Primary sub-categories: a dimension×stage cell can have units in it and still miss the stage's real concern if none of those units satisfy a Primary sub-category.

Using it for adopter guidance: the core question per dimension×stage is the agent's opening probe with a live adopter. If a Primary sub-category for their current stage is unaddressed, that's what the agent asks about next - surfacing relevant corpus units as it goes.

3. The pathway document - output structure

The pathway document - output structure

A pathway document is not a case study. A case study documents what was built. A pathway document is written for the next adopter - what they would need to decide, the alternatives they would consider, the conditions under which different choices are correct. *Physical analogy: a pathway is not the route the pioneer took. It is the marked trail they left for the next traveller.*

#	Section	Purpose	Contents
0	Reading guide	Orients the adopter	What a pathway is. How retrieval works. Where reusable value concentrates. How to navigate by dimension/stage.
1	Pathway identity	Names the deployment for retrieval	Deployment name, sector, geography, population served, stage reached, contributing organisation, key dates, 2-sentence summary, Scale/impact achieved - Headline usage and outcome numbers, as-of date
2	Effort details	Details of cost, time and effort	1. Cost anchor - Setup + run-rate cost order of magnitude, as-of date 2. Build effort - Time, team size, partner count to reach current stage
	Downstream Adoptions	Known downstream adopters / reuse record	Who has since built on this pathway, and how much faster - pathway-level metadata, not a tagged unit
2	The 4×4 grid	Shows where knowledge is dense and where gaps remain	Coverage map: 4 dimensions × 4 stages, density symbols (●●● / ●● / ● / ○). Empty/thin cells checked against Primary sub-categories, not raw count.

3	Micro-innovations	The core reusable content	Tagged units organised by dimension, then stage. Each unit: decision, alternative considered, reason for decision, condition tag (applies when / fails when), before→after outcome,
4	Toolkits and playbooks	Reusable artefacts and process knowledge	Technical templates, governance frameworks, testing protocols, prompt patterns, vendor criteria - each tagged with the purpose and conditions for reuse.
5	Retrieval guide	Helps the next adopter find what's relevant fast	Organised by adopter intent (e.g. "I need to avoid vendor lock-in at Define stage") → points to relevant units and toolkit assets.
	Source Trace - appendix (contributor-only - never adopter-facing)	Traces what was used to build the pathway, for future reconciliation	Table keyed by source file → which Sections/fields/units it covers, and whether it's primary or confirms-only

Worked example - a single micro-innovation unit:

Failure: Direct hardwiring of the AI layer to the ICAR database - a backend change required an AI-layer rebuild.

Fix: Separated the AI layer from the data layer using a standardised API gateway. The AI system retrieves data but does not own it.

Insight: At scale with multiple data sources, this is the difference between a maintainable system and a fragile one.

Condition - applies when: Multiple data sources with different owners and update cadences; government deployment where data accountability must remain with named departments.

Condition - fails when: Single, stable, internally-owned data source with no requirement for departmental accountability separation.

Before → After: Before: data errors required rebuilding the bot prompt architecture. After: data errors are fixed by the data owner without touching the AI layer.

4. Extracting units from raw material

Whether the raw material is an interview transcript or a document dump, the same discipline applies before anything gets treated as a unit:

Step	What it means
Tag every unit	Dimension + Sub-category + Stage + Type + Condition tag. No untagged units enter the corpus. Also relevant at:[Stage, Stage] - optional. Captures where a unit is <i>useful</i> beyond its stage of origin. Does not count toward the Section 2 coverage grid - grid density is driven by Stage (origin) alone, so gap-detection stays honest. This field feeds only the Section 6 retrieval guide and adopter-facing navigation. Stage reflects where the evidence was discovered (counted in the grid). "Also relevant at" reflects where it's useful (not counted, used for retrieval only). Never merge the two.
Write the before→after	Every tactical/strategic unit needs an outcome statement. Without it, it's a lesson, not a finding.
Name the failure specifically	"It didn't work" is not a unit. Name the failure, the fix, and the threshold or insight the fix revealed.
Flag the gaps	Check against Primary sub-categories per stage, not raw cell density - a filled cell can still miss its stage's real concern.
Don't fabricate	If a before→after, a named individual, or a condition isn't in the source, write "Not documented in the source" rather than inventing it.

The synthesis test: could someone who never saw the raw material make a different decision because of this unit? If yes, it's a real unit. If it just describes what happened, it isn't.

Appendix Source Trace - spec

Appendix - Source Trace (*contributor-only; never surfaced in an adopter-facing AI-layer response, in any mode*)

A table keyed by source file, not by unit:

Source file	Covers	Notes
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[filename/doc, as-of date]	[Sections/fields/unit ranges populated from it]	[Primary source / confirms only / superseded by newer file, etc.]
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Rules:

- Key by raw source file, not by content item - scales better than a per-unit or per-field row.
- "Covers" should reference actual Section numbers and unit ranges (e.g., "Pathway Identity - all fields; Units 1–20; Toolkits table; Problem→Solution patterns"), not vague descriptions.
- When a file is derivative of another (a summary, an earlier draft), say so explicitly and mark it "confirms, doesn't add" rather than listing it as an independent source for the same content.
- This appendix sits outside the Section 0–6 count - it's not part of the adopter-facing structure.